

Information: Study from Sequence Analysis

The meta-alphabet



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Every text message, DNA strand, computer file, or digital image can be stored in an ordered list of symbols; whether binary bits, decimal digits, hexadecimal letters, RGBA pixels, nitrogenous bases, we can appreciate how information, although expressed differently, can be **digitized into a sequence of different “alphabets”**.

While the alphabet and length may differ, the information is stored in a unique sequence. So, it is not mandatory to use a specific alphabet.

Metalphabet would refer to the Set of all alphabets.

In this context **‘alphabet’ is just a set of symbols**; (0,1) if binary, (0,1,2,...,9) if decimal, (a,b,c,d,...,z) if the classic alphabet, (⟨б⟩, ⟨в⟩, ⟨г⟩, ⟨д⟩, ⟨ж⟩, ...) if Russian alphabet, (ひ, ら, が, な, ...) if Japanese.

Notice that **an ‘alphabet’ can also be pixels**, nitrogenous bases in DNA, etc. Symbols are free of interpretation; they could be anything. The important part is that they will be unique; if two symbols are not

exceptional, then they are the same or correlated. For alphabets we initially want independent symbols, although it is possible if they are dependents of each other, yet as a first approach, independent symbols.

The idea of this story was to go beyond the original definition of why entropy is formulated the way it is, what it really represents, and it was too long, so split for the next story:

The Shannon Entropy Generalization: A New Definition for Entropy
Shannon's entropy is heuristic; it works well for systems with a large number of permutations, but it is less effective...

medium.com

Key idea: the meaning of a message does not depend on the particular symbols chosen to represent it (the alphabet used). By treating every possible symbol set as part of a *meta-alphabet*, we can analyse communication without fixing any specific alphabet.

Indexing: Sequences as carriers of information

In any communication system, whether it's a text message, DNA strand, computer file, or digital image composed of pixels, we work with sequences: **ordered** lists of symbols where **the order matters**; swapping two digits changes the meaning entirely.

Each arrangement (a *permutation*, not a simple combination) produces a different message, even if using the same symbols.

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ABC vs. CBA
message 1 message 2

This is important because if we want to represent 8 different states (or messages) in a sequence where the order does not matter, then we will need 8 different unique symbols, while if the order does matter, we can encode the same information in a sequence of $\log(8)$ symbols (log base-2 if bits).
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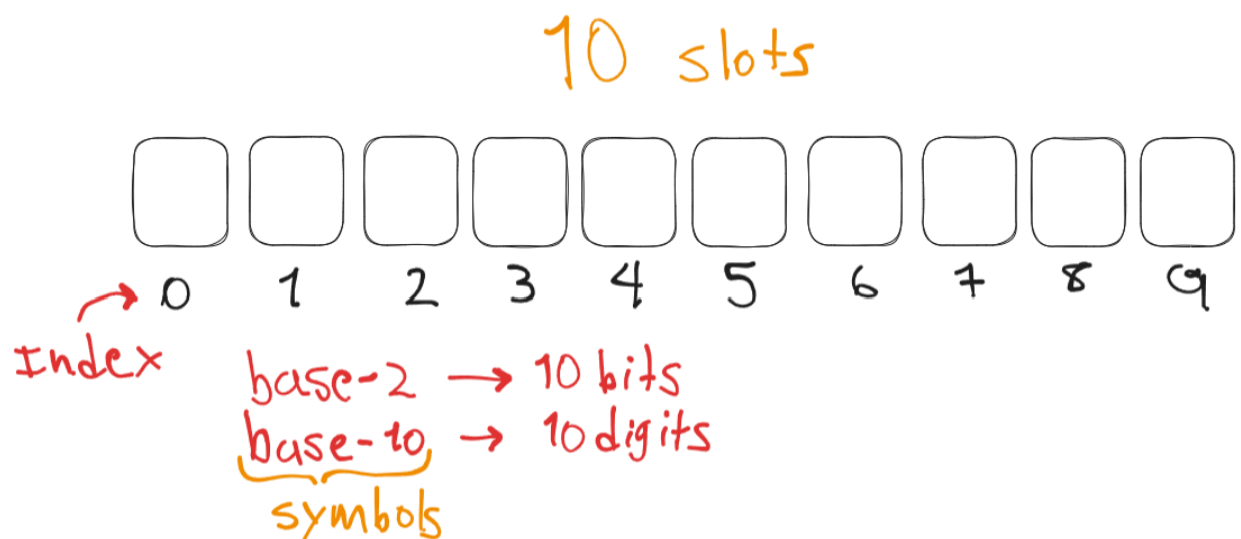
indexing {
000 → A
001 → B
010 → C
011 → D
100 → E
101 → F
110 → G
111 → H
8 symbols
2 symbols
+ indexing

So, **key idea**: If we deliberately choose the order to matter, we can encode the same information with fewer symbols by adding an index which represent the order.

Sequence + Alphabet, Glossary

Alphabet here is just a collection of symbols whose purpose is to be used in a sequence. Data will be stored in an alphabet permutation, glossary concepts:

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- Slots (D): The number of positions in the string. In binary each slot is one *bit*; in decimal each slot is one *digit*; in hexadecimal each slot holds one hex-digit.
- Index (d) of the slot: A counter that tells us which slot we're looking at (1, 2, 3,...). Because order matters, keeping track of indices lets us rearrange or reference any symbol without confusion.
- Symbols (b): The size of the alphabet from which symbols are drawn: 2 for binary (0 / 1), 10 for decimal (0–9), 16 for hex (0–F).

- The Greater number is N: The largest number that can be represented with d slots in base- b .
- Configuration or States (total number is $N+1$, because of 0): the different ways a sequence can exist.

Under this context, **information** is the cardinality or number of elements of the permutations. Example: the information of 3 bits is 8 possible permutations; 4 bits have more information because 4bits can represent 16 permutations. Here information is just counting the number of elements in a set. Yet it can be considered as “meaningless information”, the semantic meaning of each permutation relies on future bijections with permutations of other Sets.

Counting Configurations (states): How Slots and Symbols Permute

The way they connect with each other in base-2 is like this:
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$$N = b^d - 1$$

biggest number total states from zero

We can calculate the summation of elements inside the set (for each iteration, index) by properties of the set (the size, cardinality).

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$$\sum_{d=0}^{D-1} b^d = b^D - 1$$

Handwritten annotations: "along index" points to the summation symbol; "D-1" is circled; "D" is circled; "out of set" points to the b^D term.

Curious coincidence:

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$$\sum_{d=1}^T b^d = b^T - b^1 \sim \int_1^T b^d db = \left. \frac{b^d}{d} \right|_{d=1}^{d=T}$$

Handwritten annotations: "T" is circled with "max" above it; "1" is circled with "min" below it.

The relation between accumulating through indexes and the boundaries might not be obvious, so let's check with a numerical **example**:

We will use 3 bits of information, so $b=2$ (bits have 2 symbols), $d=3$ (3 slots sequence, 3 bits). That is enough to find the number of configurations 2^3 , also the last number 2^3-1 :

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$$7 = 2^3 - 1$$

(3) → number of bits
 2 → base in bits
 1 → zero offset
 biggest number → 7
 → $7 \equiv 111 \pmod{2}$

What about the Summation? If we want to calculate the same number through the index:

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$$\sum_{n=0}^3 2^n = 1 + 2 + 4$$

(3) → index
 4 → $2^4 - 1$

The reason is simple, you can add 1–1, and **play 2048**:

$$\begin{array}{c}
 \begin{array}{ccccccc}
 & 2^0 & & 2^1 & & 2^2 & \\
 1 & + 1 & + 2 & + 4 & - 1 \\
 \hline
 & 2^{2^1} & + 2^{2^1} & + 4^{2^2} & - 1 \\
 \hline
 & & 4^{2^2} & + 4^{2^2} & - 1 \\
 \hline
 & & & 8^{2^3} & - 1
 \end{array}
 \end{array}$$

Alternatively, you can check this using geometric series, which generalizes to other bases:

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$$\sum_{d=0}^{D-1} b^d = \frac{b^D - 1}{b - 1}$$

What changes is that the last 1-digit number is now (b-1) and not “1”.
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$$\sum_{d=0}^3 10^d = \frac{10^4 - 1}{10 - 1}$$

↓

$$10^0 + 10^1 + 10^2 + 10^3 = \frac{10^4 - 1}{10 - 1}$$

↓

$$\underbrace{9 \cdot 10^0 + 9 \cdot 10^1 + 9 \cdot 10^2 + 9 \cdot 10^3}_{9999} = \underbrace{10^4 - 1}_{10000 - 1}$$

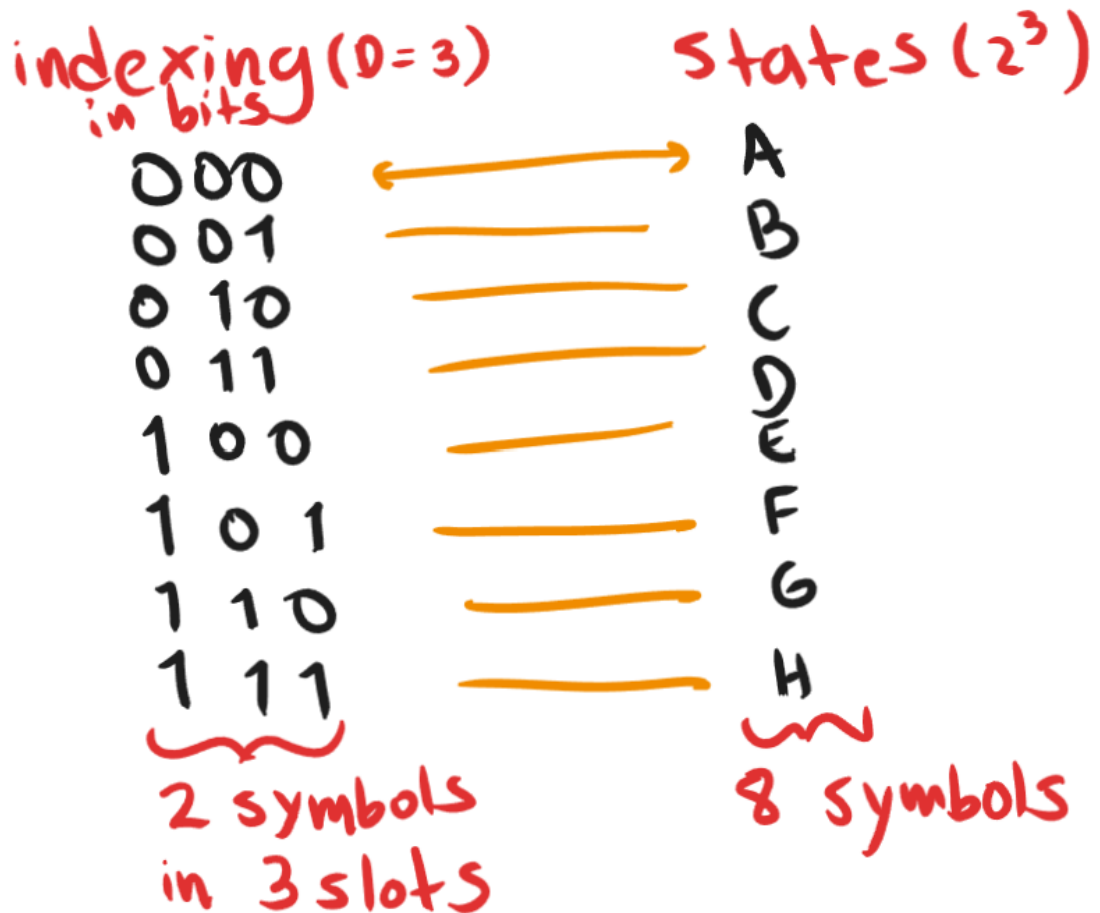
↪ last number in 4 digits ↪ first number 5 digits - 1

This works for all different bases.

Morphisms; cross alphabetization

We focus on morphisms rather than on the alphabet itself; bijections, in particular, let us track information while preserving data along transformations.

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In the above example, we can do 2^3 states, so we can map each of them to a message, and each specific sequence to rename or point to another message.

Example: 000 means "hello" and 111 means "bye", and we can map every number from 0 to 2^8-1 to a different message.

We can do this in different alphabets, not only with two symbols for binary numbers. What is important to keep from this is that the number of messages that can be encoded in an alphabet depends on the number of permutations of the sequence.

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$$\text{messages} := \{ \text{ABC}, \text{ABB}, \text{CBC}, \dots \}$$

(symbols, length)

$$\text{len}(\text{messages}) = \overset{1}{3} \times \overset{2}{3} \times \overset{3}{3} \times \overset{4}{3} \rightarrow 4 \text{ slots}$$

$\Rightarrow 3^4$ $\hookrightarrow 3$ possible symbols in each slot

The cross-alphabetizations come from the ability to write the same messages in different alphabets:
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$$\text{messages} := \{ \text{ABC}, \text{ABB}, \text{CBC}, \dots \}$$

($\{A, B, C\}$, 4)
 3 symbols 4 slots $\Rightarrow \text{len}() = 3^4 =$

$$\text{messages} := \{ 0000..1, 101..0, 111..0, \dots \}$$

($\{0, 1\}$, 7)
 2 symbols 7 slots

As long as two sets have the same size, we can establish a 1-to-1 connection, called a bijection. Yet if one is bigger than the other, we can artificially add empty messages and make it work.
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$$\text{semantic messages} = \text{permutation} * \text{Alphabet}$$

One last thing I wish you to keep from this is, think of semantic messages as independent of the syntax, meaning we can write the same meaning in different symbols (bits, pixels, digits, hex, nitrogenous bases, etc.). But also, we are used to thinking of “message” as the specific syntax, so let’s bifurcate the concept into two: semantic message and syntax message.

In a semantic message, we can change the alphabet, and all we need to talk about the same semantic message in a different alphabet is using the respective different permutation. Also is possible to mean different semantics by using the same syntax, but with a different alphabet. (example: “Pie” in English or Spanish).

Entropy comes in when the order is lost

Entropy is strongly related to randomness; if swap randomness didn’t exist in a sequence, then entropy would be an unusual concept. **Entropy’s relevance is tied to the statistical consequences of randomness.**

This randomness is tied to **the “informational” uncertainty principle** and the fact that nature is intrinsically random.

My observation: The Universe randomly swaps positions in the syntax of a sequence. When it happens enough times, the index stops making sense:

When the index ‘n’ lacks sense, you can expect this in nature as a mixing of order (random movement).

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$$\begin{aligned}
 &\text{original} \\
 &\text{message}_1 = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9] \\
 &\text{swap}^1 * \text{message}_1 = [3, 1, 2, 0, 4, 5, 6, 7, 8, 9] \\
 &\text{swap}^2 * \text{message}_1 = [3, 1, 2, 0, 6, 5, 4, 7, 8, 9] \\
 &\text{swap}^n * \quad \quad = \text{random}
 \end{aligned}$$

If the indexes get lost, do we lose information? **Not all, but yes**, we lost a lot of information to share. **But that is exactly how the Universe works**, and that is what the second law of thermodynamics means.

Now all that matters is the amount of a bit turned on, or off, not the order in which they are. In mathematics, this is what Summations does. I mean, even if randomness does a mess in the order, the total elements remain the same; this allows the **conservation of Energy** across swaps (not classic energy, but informational, building this concept...).

The addition has a fundamental property: the order doesn't matter. $a+b=b+a$, this is commutativity. Commutativity is the backbone of order-lost information. This applies once the order is lost in the sequence, but yet energy is conserved, if we split the sequence in half, we know we can add them up and give the same energy without caring about the order.

What I mean is:

1. We have a syntax message.

2. The swap principle acts (related to the “informational” uncertainty principle).
3. We lose a kind of information along random swaps, **BUT there are other kind of information that never lose.**
4. The information we don't lose will be called Energy.

Abstractly, we can build different definitions of energy here, and only one of them will be the identical one with physics energy. (So, until now, the same word, different meaning, but think of this as an abstraction ~ generalization, then physics energy would be a specific case of this definition of energy).

Up to this point, the definition of Energy is whatever remains constant during random swaps.

Example 1

For 3 bits, this means 100, 010, and 001 are equivalent under disorder rules; if we keep in mind that the order matters, they are not. If the order does not matter, we can add the bits as a definition of Energy and find out that Energy is preserved under random swaps: $1+0+0=0+1+0=0+0+1$

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original
 message₁ = 001
 ↪ swaps messages = {001, 010, 100}

Energy
 Measure = sum (symbols)
 0+0+1
 = 1

Here, the definition of energy is just added up every symbol in the sequence, personally I like more convenient for physics to use the self-dot product, but would detail it on future story.

Addition has a commutative property that allows us to build energy measures easily, as the order of addition doesn't matter.

Combinatorics and permutation

Using combinatorics, we know we can count the number of permutations that have the same number of symbols, the same energy:

As summation removes any notion of order, we can simulate these operations using the binomial theorem (two symbols followed by the expansion)

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$$(\boxed{A} + \boxed{B})^3 = (A + B)(A + B)(A + B)$$

2 symbols
3 slots

$$\rightarrow \sum_{n=0}^3 \binom{3}{n} \underbrace{A^{3-n}}_{\text{count of } A} \cdot \underbrace{B^n}_{\text{Energy of states}}$$

The combination counts the number of a given sort of symbols, for example, if we want to know how many configurations there are when there is 1 once, and 0 twice, it will be this term:

$$\binom{3}{1} \underbrace{A^2}_{\text{twice } A} \underbrace{B^1}_{\text{once } B}$$

count

So, we connected the loss of order with combinatorics (from the binomial theorem).

What happens here with the messages? Now we have all zeros, 1 zero in total, 2 zeros in total, 0 zeros in total, 4 messages in total. When the order was in place, there were 8 configurations, and now, there are 4.

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$$(1+x)^3 = 1 + 3x + 3x^2 + x^3$$

$\underbrace{\quad}_{\substack{\text{A} \quad \text{B}}} \quad \underbrace{\quad}_{\substack{\text{A}^3 \text{B}^0 \quad \text{A}^2 \text{B}^1 \quad \text{A}^1 \text{B}^2 \quad \text{A}^0 \text{B}^3}}$
 $2^8 \text{ states vs. } 4 \text{ Energy states}$

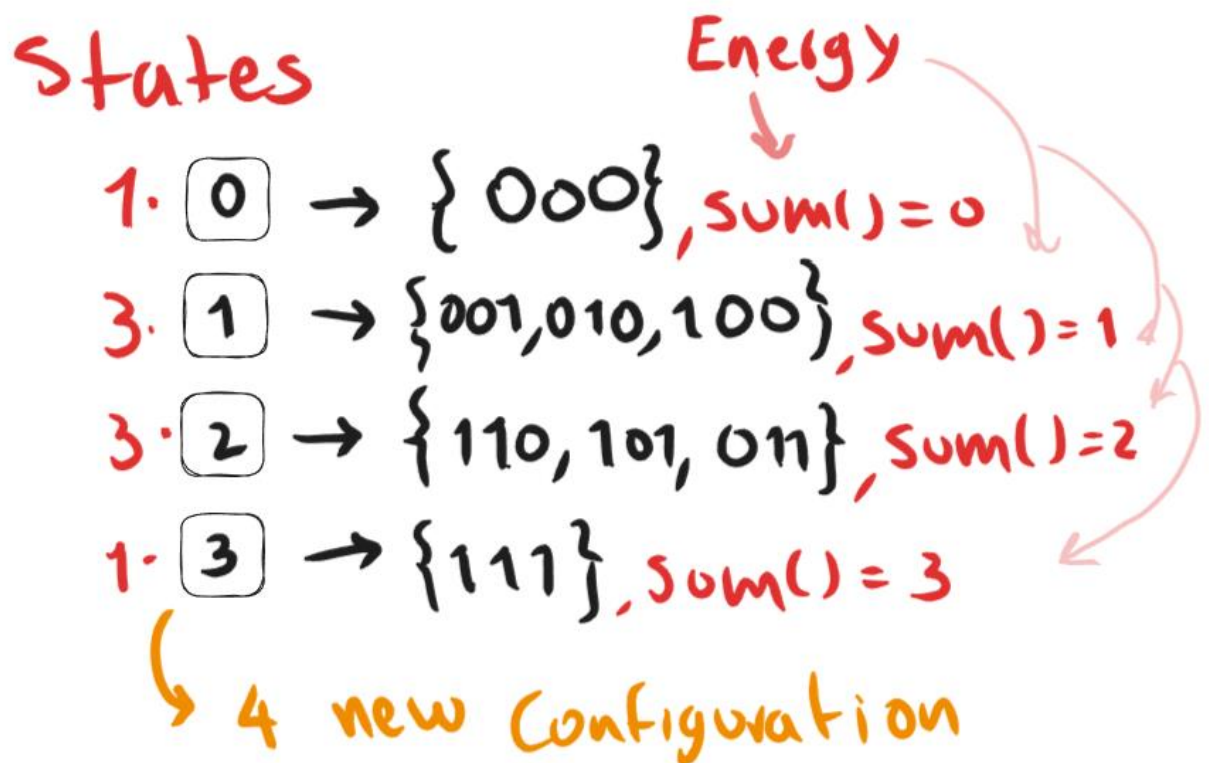
The number of configurations after the order is lost is the number of slots+1.
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$$(1+1)^3 = \sum_{n=0}^3 \binom{3}{n} * \text{Energy states}$$

$\underbrace{(1+x)^3}_{2^3=8} \quad \text{vs} \rightarrow 4 \text{ terms}$

If we try to analyse how the counting is working for the same energy configuration:

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When it is a multinomial, we get $\text{comb}(D+b-1, b-1)$ terms:
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$$\{a, b, c, \dots\}^D \Rightarrow \binom{D+b-1}{b-1} \text{ terms}$$

It follows counting all possibilities of $a+b+c+\dots=D$, for $\{a, b, c, \dots\}^*$ positive integers. The related energy comes from assigning each symbol a weight such that they are independent for linear integers combinations. So, prime numbers. Here doesn't care the quantity, but the quality; being different, not bigger.

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$$\{a, b, c, \dots\}^D \Rightarrow D! \sum \frac{a^i b^j c^k \dots}{i! j! k! \dots}$$

number of terms ↑ $i+j+k=D$

Recapitulation

Initially, the position of each symbol in a sequence is crucial: swapping two elements changes the message entirely. When random swaps are allowed, this positional information disappears, but there is also information that does not disappear, which will be called **Energy**.

There is a law in nature that randomly swaps symbols; it's related to the uncertainty principle. So, we want to understand this process, what is preserved, and what is lost.

At first, it reduces the cardinality, the number of different states: For a binary alphabet, the set of distinct states collapses from b^d to just $d+1$; for larger alphabets, we must use the multinomial theorem instead of the binomial one.

Objectively, the number of states is the same, but **subject to an energy measure**, it is reduced. Some states might be repeated, and next, we would say they are more common than others. (oooooooo) vs. (o,ooo,ooo,o) both with 8 'o', what changes is interpretation of subgroups.

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$$1 + 3 + 3 + 1 \rightarrow 8 \text{ non unique states}$$

#1 #2 #3 #4
 4 unique states

I didn't use "microstates" and "macrostates" until now because these concepts are more related to thermodynamics, but basically, the generalization Energy I wrote above is a macrostate measure, while each permutation of the symbols is a microstate.

Macrostate measures provide information about all microstates without treating any of them preferentially, so random swap invariants, which make it impossible to guess the microstate from it.
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$$\text{Macro State} = \sum_{n=0}^{N-1} f(\text{microstate}_n)$$

One example of a macro-state measure is energy, just as we saw before.

For calculation facilities, we prefer definitions of energy which directly connect information along the micro-states with the boundaries of the macro-state. This is the case of geometric series and analytical functions,

because this provides factorization properties, which provide a richer flexible algebra.

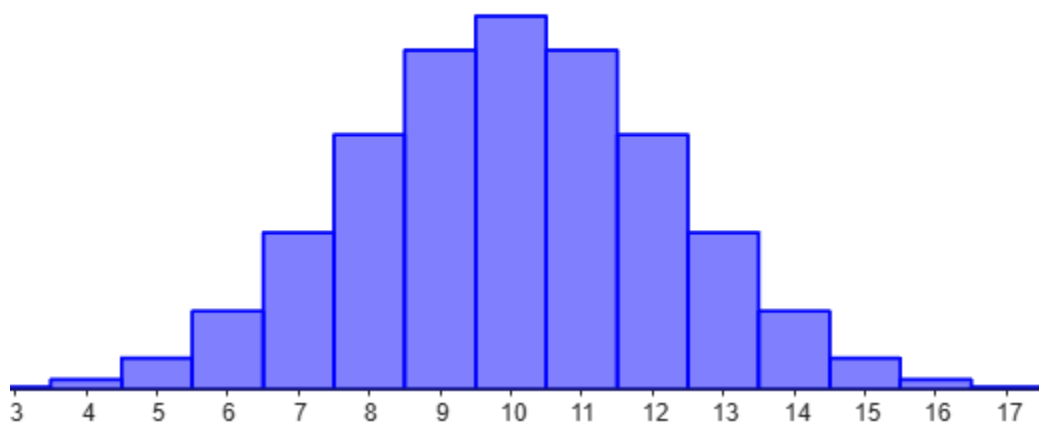
Relation with truth and logic

Information lies not in syntax related to a respective alphabet; instead, information is cross-alphabetic, and semantic messages are meta-alphabetic. What really matters is not the study of the alphabetic set, but the **morphism** between different alphabetic sets, morphism along the meta-alphabet.

In logic, we would say something is true if there is a morphism between the permutation inside an alphabet and another, the first set represents the “representation” and the second set the “reality”. When the morphism between them exists, then the “representation” is said to be true under the “reality” set.

While syntax lives inside an alphabetic set, semantics lives in the morphism of different alphabet sets (meta-alphabet).

The loss of order makes some configurations far more common than others; this is how entropy is born. The most frequent arrangements lie near the center (e.g., [1,3,3,1] for a 3-length 2-symbol alphabet). Their frequencies follow a binomial law that approaches a Gaussian curve as the number of slots d grows. They come from combinatoric rules from the loss of indexing.



What does this imply? When we observe a macro-state with a given energy, the most likely internal configuration is the index central configuration, its probability is highest, whereas configurations at the extremes occur with essentially zero chance.

With 3 bits only, the probability of all zeros or all ones is $1/2^3$, while the probability of 1 zero is $3/2^3$, (dividing by the total number of configurations)
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$$\frac{(1+x)^3}{(1+1)^3} = \frac{(1+x)^3}{2^3} \rightarrow \text{normalization}$$

This is called normalization (there are many other types of normalization), and allows us to study counting of states as probability, as # positives_cases/#total_cases.
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$$(1+x)^3 = \underbrace{\frac{1}{8}}_{\text{low probable}} + \underbrace{\frac{3}{8}x + \frac{3}{8}x^2}_{\text{High probable}} + \underbrace{\frac{1}{8}x^3}_{\text{low probable}}$$

(errata: division by 8 should be on both sides)

From here, we can relate it to the generative function for the binomial distribution:

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$$(p + q)^D = \sum_{d=0}^D \binom{D}{d} p^d q^{(D-d)}$$

This is the foundation of entropy. We usually maximize entropy by looking for the microstate that repeats the most. For big systems, the most repeated microstate is just the one with 99.9999% probability to be, the one in the middle of a binomial expansion.

Last recapitulation:

Information is usually not accurately defined; it lends itself to confusion.

Then, allow me to define it as:

- Information is the cardinality of the permutations of a sequence. The number of different permutations a sequence can have. (*This is how we think of information in computers*)

Think of it as a Disk drive or a USB, even if it is “empty”, we still say it has 16GB. So, information doesn’t mean it has any semantic message on it; it can be empty, yet the information capacity is the same. **“Meaning” can be subjective, so I think it is better to define information objectively as the capacity of the number of permutations.**

- Semantic meaning: (*This is how we use to think of information in language*), when a message carries “information”, what we mean is that it has a bijection with another set, the “meaning” is restricted to the bijection. Without a bijection, there is no meaning at all.

16GB of a USB can be empty, or not. Whether the permutation of its sequence means something or not depends on a decoder, decryption, or is mathematically equivalent to a bijection. So “empty” is really subject to the decryption used. 000000000000 can mean nothing or can mean a crucial signal for anything. Again, the context of whatever syntax means relies on a bijection rule, so the semantic meaning of information is a morphism rather than a specific permutation. Then, meaning is cross-alphabetic, meta-alphabetic.

Energy would be a measure that partially summarizes all microstates, meta-alphabetic. While entropy simply counts the number of different permutations given an amount of energy. In physics, we are especially interested in maximizing entropy because it tells us which permutations are more repeated and therefore more probable to occur.

Thanks for reading!

Mathematics

Entropy